Exelon Generation.

10CFR 50.73

CCN: 18-131

December 21, 2018

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Unit 3
Renewed Facility Operating License No. DPR-56
NRC Docket No. 50-278

Subject: Licensee Event Report (LER) 3-18-002, Revision 1

Enclosed is a revision to LER 3-18-002 to identify a second violation of Technical Specifications (TS) that occurred during a TS required plant shutdown. Revised sections are identified by revision bars in the right margin. In accordance with NEI 99-04, the regulatory commitment contained in this correspondence is to restore compliance with the regulations. The specific methods that have been planned to restore and maintain compliance are discussed in the LER. If you have any questions or require additional information, please do not hesitate to contact Matt Retzer at 717-456-4351.

Sincerely,

Matthew J. Herr
Plant Manager
Peach Bottom Atomic Power Station

MJH/dnd/IR 4175898

Enclosure

cc: US NRC, Administrator, Region I
US NRC, Senior Resident Inspector
R. R. Janati, Commonwealth of Pennsylvania
D. Tancabel, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs

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NRR

NRC FORM 366 (04-2018)

U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
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4. Title

Reactor Coolant System Pressure Boundary Leakage Resulting in Technical Specification Required Shutdown

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number 05000
09	21	2018	2018	- 002	- 1	12	12	2018	Facility Name	Docket Number 05000

9. Operating Mode: 1

10. Power Level: 100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☒ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☒ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Matthew E. Retzer, Regulatory Assurance Manager	717-456-4351

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
B	BJ	PSP	NA	Yes					

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No				

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On 9/21/18, the High Pressure Coolant Injection (HPCI) system was declared inoperable due to two inoperable differential pressure indicating switches that are used for isolating HPCI in the event of a break in the HPCI turbine steam line. A containment entry was made and it was determined that the instruments were incapable of performing their design function due to a leak on a one-inch diameter pressure sensing line. The line is a reactor coolant system pressure boundary, resulting in a Technical Specification required shutdown in order to repair.

The cause was determined to be vibration-induced fretting of the instrument line by a support bracket. The section of the pipe with the leak was replaced.

There were no actual safety consequences as a result of this event.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
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Unit Conditions Prior to the Event

Unit 3 was operating in Mode 1 at approximately 100% rated thermal power. There were no structures, systems or components out of service that contributed to this event.

Description of Event

On 9/20/18 at approximately 1930 hours, an Equipment Operator, during normal rounds, identified that differential pressure switches DPIS-3-23-076 and DPIS-3-23-077 [EIIS:PDS] for the High Pressure Coolant Injection (HPCI) system [EIIS:BJ] were reading lower than expected.

Upon further review and investigation, it was determined that the switches would not have been able to perform their Technical Specification (TS) 3.3.6.1 required function of isolating the HPCI steam line in the event of a break. The two switches were declared inoperable on 9/21/18 at approximately 1755 hours and were inoperable for approximately 22-1/2 hours as a result of this event. Required actions of TS 3.3.6.1 were performed, including isolation of HPCI steam line. As a result of the isolation, TS 3.5.1 Condition C was entered for HPCI being inoperable, which requires the system to be restored within 14 days.

Further analysis of the low differential pressure indicated on the DPISs suggested a possible leak of the instrument line upstream of the DPISs. This was supported by a small increase in measured drywell leakage that was identified on 9/20/18. A reactor power reduction to 35% was initiated on 9/22/18 at 0200 hours to support an entry into the drywell. The drywell inspection verified leakage of the 1-inch diameter instrument line (Schedule 80, Type 304 stainless steel), which is a reactor coolant system pressure boundary. TS 3.4.4 Condition C was entered at 0955 hours on 9/22/18, which requires the unit to be in Mode 3 in 12 hours and Mode 4 in 36 hours. A controlled shutdown was initiated and the unit entered Mode 3 at 2029 hours on 9/22/18 and Mode 4 at 0701 hours on 9/23/18.

The leak was located at a point where the instrument line was supported by a rectangular shaped bracket. Vibration had caused the bracket to wear the instrument line, eventually resulting in a through-wall steam leak. The section of pipe that contained the leak was replaced. Similar small-bore piping was inspected and repairs performed as necessary.

Following repairs, a normal startup was commenced and Unit 3 entered Mode 1 at 1638 hours on 9/25/18.

Analysis of Event

This event is being reported in accordance with the following:

- 10 CFR 50.73(a)(2)(i)(A) – the completion of any nuclear plant shutdown required by the plant's Technical Specifications. TS 3.4.4 Condition C requires the unit to be in Mode 3 within 12 hours for any reactor coolant system pressure boundary leakage. This TS action was entered on 9/22/18 at

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0955 hours when leakage was identified on an instrument line associated with monitoring HPCI steam line differential pressure. The instrument line is part of the RCS pressure boundary.

- 10 CFR 50.73(a)(2)(i)(B) – any condition prohibited by plant Technical Specifications. Two conditions prohibited by TS existed:
  1) Differential pressure switches DPIS-3-23-076 and DPIS-3-23-077 were identified as reading low on 9/20/18 at approximately 1930 hours. Further analysis determined that they would not have been able to perform their safety function of isolating the HPCI turbine steam line in the event of a steam line break. TS 3.3.6.1 Condition B requires the isolation capability performed by these switches to be restored in one hour. If not restored within one hour, Condition F requires the penetration flow path to be isolated within one hour (two hours after discovery). Because of the additional analysis performed to determine if they could perform their safety function, the required actions were not performed until 9/21/18 at approximately 1755 hours, which is approximately 22-1/2 hours after discovery.
  2) Differential pressure switches DPIS-3-23-076 and DPIS-3-23-077 were identified as reading low on 9/20/18 at approximately 1930 hours. This was not known to be due to pressure boundary leakage until 9/22/18 at 0955 hours, at which time TS 3.4.4 Condition C was entered. TS 3.4.4 Condition C requires the unit to be in Mode 3 in 12 hours and Mode 4 in 36 hours for any pressure boundary leakage. The unit entered Mode 3 at 2029 hours on 9/22/18 and Mode 4 at 0701 hours on 9/23/18. Because there is firm evidence that pressure boundary leakage existed prior to the time of discovery on 9/22/18, the TS required completion times were not met. Mode 3 was entered approximately 49 hours and Mode 4 was entered approximately 59 hours from the time low readings were identified on the DPISs.
- 10 CFR 50.73(a)(2)(ii)(A) – the condition of the plant, including its principal safety barriers, being seriously degraded. Leakage was identified on an instrument line associated with monitoring HPCI steam line differential pressure. The instrument line is part of the RCS pressure boundary, which is one of the principal safety barriers.
- 10 CFR 50.73(a)(2)(v)(D) – any condition that could have prevented the fulfillment of the safety function of a system needed to mitigate the consequences of an accident. HPCI is considered a single train system and was required by TS to be declared inoperable as a result of this event.

Cause of the Event

The one-inch diameter instrument line was aligned such that it was in contact with one side of a rectangular support. Vibration-induced wear (fretting) reduced the wall thickness of the pipe at the contact point until a steam leak developed. It could not be determined why the pipe was aligned such that it was in contact with the support bracket.

Corrective Actions

The section of pipe where the leak occurred was replaced. Inspections were performed on other small-bore piping in the drywell that are supported by the same style support bracket and subject to vibration. One other location was identified where fretting was significant enough to proactively have the pipe replaced.

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Piping with less significant fretting was identified at nine other locations. These wear areas were wrapped with a layer of stainless steel sheet metal to protect the line until permanent repairs can be completed if needed. Additional corrective actions are documented in the corrective action program.

Previous Similar Occurrences

No previous similar occurrences have been identified of RCS pressure boundary leakage due to fretting. Unit 3 LER 17-001 documents RCS pressure boundary leakage due to a weld failure in a one-inch diameter instrument line. Unit 3 LER 05-003 documents RCS pressure boundary leakage due to a weld failure in a one-inch diameter equalizing line in the Residual Heat Removal System.